

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Semester Examination – January 2023

Course: B. Tech.

Branch : CE / CSE / CS

Semester : V

Subject Code & Name: Machine Learning [BTCOC503]

Max Marks: 60

Date:

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/ CO)	Marks
Q.1 Solve the following questions.		[12]
A) Differentiate between Supervised Unsupervised Machine Learning approaches with suitable example.		2
B) How do you choose best split in the Decision Tree algorithm		2
C) Write Short note on: (a) Cross Validation (b) Inductive bias		2
Q.2 Attempt the following questions.		[12]
A) Write a note on Bayes theorem and illustrate maximum likelihood method for predicting probabilities in Bayesian learning with an example.		2
B) What is instance based learning? Explain importance of feature reduction while solving problem using machine learning technique.		1
Q.3 Solve Any Two of the following.		[12]
A) How to find right Hyperplane in Support Vector Machine explain with suitable Example.		2
B) Explain Logistic Regression on the basis of following points: (a) Features of Logistic regression (b) Types of Logistic Regression.		2
C) Differentiate between Multinomial Logistic Regression and Binomial Logistic Regression.		2
Q.4 Solve any TWO questions of the following.		[12]
A) Differentiate between machine learning and Deep Learning.		2
B) Explain basic architecture of multi layer network. Explain back-propagation, in multi layer network.		1

- C)** Write note on: 1
1. PAC Learning Model
 2. Ensemble Learning

Q.5 Solve Any Two of the following. **[12]**

- A)** How does k-means clustering algorithm works? 2
- B)** How to choose optimal value of k in k-means clustering technique? 1
- C)** Write note on: 2
- (a) adaptive hierarchical clustering
 - (b) Gaussian mixture model

***** End *****